

Jack Lu

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Education

New York University

Sep 2023 – Present

Ph.D. in Computer Science (3.97/4.00 GPA)

Advised by: Mengye Ren, thesis committee: Greg Durrett, Kyunghyun Cho

Research focus: Self-Improvement, Reasoning, World Models

University of Waterloo

Sep 2018 – May 2023

B.Math. in Computer Science and Mathematics (4.00/4.00 GPA)

Advised by: Raquel Urtasun, Sanja Fidler, Alexander Wong

Research focus: Generative Modeling, Autonomous Driving, ML for Health

Selected Publications

TacitVLA: Reasoning Vision-Language-Action Models Without Reasoning Latency

Under review

Jack Lu, Yurong You, Boyi Li, Yulong Cao, Thomas Tian, Chaowei Xiao, Xiangyu Chen, Wenjie Luo, Boris Ivanovic, Marco Pavone, Yan Wang

The Surprising Effectiveness of Deleting Weights in LLM Post-Training

ICMLW FoGen 2026, NeurIPS 2026

Jack Lu*, Zhenbang Yang*, Mike Lasby, Tejas Pote, Yani Ioannou, Mengye Ren

HyperThink: Text-to-Parameter Hypernetworks for Efficient Reasoning

COLM 2026

Donggyun Kim*, **Jack Lu***, Chanwoo Kim, Mengye Ren, Seunghoon Hong

When Does Verification Pay Off? A Closer Look at LLMs as Solution Verifiers

ICLRW RSI 2026

Jack Lu*, Ryan Teehan*, Jinran Jin, Mengye Ren

Context Tuning for In-Context Optimization

ICML 2026

Jack Lu, Ryan Teehan, Zhenbang Yang, Mengye Ren

Solaris: Building a Multiplayer Video World Model in Minecraft

NeurIPS 2026 (Spotlight)

Georgy Savva*, Oscar Michel*, Daohan Lu*, Suppakit Waiwitlikhit, Timothy Meehan, Dhairya Mishra, Srivats Poddar, **Jack Lu**, Saining Xie

SkillFactory: Self-Distillation For Learning Cognitive Behaviors

ICLR 2026

Zayne Sprague, **Jack Lu**, Manya Wadhwa, Sedrick Keh, Mengye Ren, Greg Durrett

ProCreate, Don't Reproduce! Propulsive Energy Diffusion for Creative Generation

ECCV 2024

Jack Lu, Ryan Teehan, Mengye Ren

SceneControl: Diffusion for Controllable Traffic Scene Generation

ICRA 2024

Jack Lu*, Kelvin Wong*, Chris Zhang, Simon Suo, Raquel Urtasun

Research Experience

Visiting Researcher – Meta

Sep 2026 – Present

Hosts: Jack Lanchantin, Jason Weston

- Researching recursive self-improvement for large language models.

Research Intern – NVIDIA

May 2026 – Sep 2026

Hosts: Yan Wang, Boris Ivanovic, Marco Pavone

- Conducted research on reasoning vision-language-action models and world models for robotics and autonomous vehicle applications.

Ph.D. Student / Research Assistant – CILVR Lab @ NYU

Sep 2023 – Present

Advisor: [Mengye Ren](#)

- Efficient adaptation: led [Context Tuning](#) (prompt optimization) and [Deleting Weights](#) (sparse post-training).
- Efficient reasoning: co-led [HyperThink](#), which turns each query into a parameter update instead of a long trace.
- Self-improvement: led a 37-model study of [when LLM verification pays off](#), and co-authored [SkillFactory](#).
- Generative models: led [ProCreate](#) (creative diffusion) and contributed to [Solaris](#) (multiplayer world model).

Research Intern – Waabi

Sep 2022 – Feb 2023

Host: [Raquel Urtasun](#)

- Achieved state-of-the-art traffic scene generation performance with diffusion models and graph neural networks (ICRA 2024).

Research Intern – NVIDIA

Sep 2021 – Mar 2022

Host: [Sanja Fidler](#)

- Applied domain-adversarial training and asset randomization methods to improve monocular 3D object detection networks.

Research Intern – DarwinAI

Sep 2020 – Dec 2020

Host: [Alexander Wong](#)

- Applied computer vision networks for CT scan analysis to detect terminal lung disease (Frontiers in Artificial Intelligence 2021).

Work Experience

Deep Learning Engineer (Intern) – NVIDIA

May 2021 – Aug 2021

- Reduced the failure rate of the autonomous-vehicle path detector by 21% with adversarial training.
- Accelerated collision detection in NVIDIA DriveSim by 7x with a custom KD-Tree algorithm.

Cognitive Software Developer (Intern) – IBM

Jan 2020 – Apr 2020

- Improved IBM Cognos Analytics chatbot's named entity recognition performance by 24% with BERT backbone.
- Developed and deployed a novel tabular data clustering algorithm with word embeddings and ontology trees.

Full Stack Developer (Intern) – Deep Trekker

May 2019 – Aug 2019

- Engineered a real-time location tracking application that enables remote robot tracking.

Honors and Awards

NSERC – Postgraduate Scholarship, Doctoral (PGS D) (\$120,000)

2024 – 2027

University of Waterloo – Winston and Diana Cherry Scholarship (\$3,000)

2023

University of Waterloo – President's Research Award (\$1,500)

2020

University of Waterloo – President's Scholarship of Distinction (\$5,000)

2019

University of Waterloo – Dean's Honours List/Distinction (all terms)

2018 – 2023

Positions of Responsibility

Teaching: Recitation Leader, Deep Learning (NYU, CSCI-GA 2572)

Advising: [Zhenbang Yang](#), [Wun Ting Chan](#), [Jinran Jin](#), [Anushka Polapally](#)

Reviewing: CVPR 2024, ICML 2025/2026, ICLR 2025/2026/2027, NeurIPS 2026, and various workshops

Skills

Languages: Python, LaTeX, C++, Scala, JavaScript, Java, R, SQL, HTML, CSS

Machine Learning: PyTorch, JAX, TensorFlow (Keras), NumPy, Matplotlib, scikit-learn, Pandas

Others: Slurm, GCP, Docker, Spark, Hadoop, Bazel, Swift